

Assignment #1

Points: 4

Deadline for submission: 27 February 2026, 23:59

Part 1 -- Realistic system:

maximum 1 page (1 point)

You are required to define a realistic system, **where the integration of 8 microservices is essential to deliver several functionalities of your system.**

In real life, you can find many examples that serve this purpose. For example, a logistics company that provides package delivery, and wants to outsource some of its business to local distributors, an airline alliance that allows any of its partner companies to book flights for its customers directly in any other company within the alliance, a hotel that wants to offer a loyalty program, which allows its members to have discounts, priority for room allocation, etc.

What is expected in this part?

In maximum one page, clearly describe:

- What your system does
- Who uses it
- What main functionalities it provides
- Why multiple microservices are necessary

You must justify that the system **cannot reasonably work as a single monolithic service** and that integration between services is required.

Notes:

- Try to keep your system **simple** but **expressive** and include only relevant information.
- **You cannot use the running example (online store) we are using on the course.**
- Each team should define **a unique system**, when you have an idea about your system register it in the shared [Excel sheet](#) so other teams do not use the same idea. You need to add your team number (provided in Moodle) and a short description of your proposed system.
- **Keep in mind your proposed system needs to be approved so when you have a clear idea about your system contact me.**

Part 2 -- Architecture:

maximum 4 pages, including diagram (3 points)

You are required to provide a **simple architecture** of your proposed system in terms of:

- Its **key microservices (8 services, i.e., 2 per team member)**
- The **dependencies between them**

What Does “Architecture” Mean in this part?

This is a logical microservices architecture, not a technical deployment diagram. Focus only on business-level service decomposition and integration. You are expected to:

- Identify 8 microservices (2 per team member).
- Clearly define the responsibility of each service.
- Show how the services depend on each other.
- Draw a simple diagram illustrating these dependencies.
- Assign which team member will be responsible for the development of each microservice (this is only for planning purposes for future assignments and does not require implementation at this stage).

In more detail, when designing your architecture, cover the following:

Step 1 — Identify 8 Microservices

Each microservice must:

- Have a meaningful name
- Have a short explanation clearly describing its business capability

Step 2 — Explain Service Responsibilities

For each service, write:

- What it does
- What data or functionality it manages
- Why it is separated from other services

The separation must make logical sense.

Step 3 — Show Dependencies

Dependencies mean:

- Which service calls another service
- Which service needs data from another service
- Which service cannot complete its work without another service

You must clearly show:

- Who depends on whom
- Why the dependency exists

Step 4 — Draw an Architecture Diagram

Your diagram must:

- Contain all 8 services
- Show arrows representing dependencies
- Be clean and readable
- Stay at business architecture level

A simple block diagram with arrows is enough. Professional UML is not required.

What is a Good Architecture?

A good architecture will show:

- Clear separation of responsibilities
- No overlapping responsibilities
- Logical integration points
- A reasonable central coordination flow
- No unnecessary complexity

Rules for assignment #1 submission

1. **February 27 is the deadline to submit your assignment via Moodle.**
2. **Submit** your assignment as a **PDF file** through **Moodle**.
3. You must submit your homework by the defined deadline, and **you will lose 0.5 points for each hour of delay.**
4. You can contact me **anytime via email or Slack (a better communication option)**, but you **need to include** all your team members in any communication.
5. **Only one submission per team is required.** Please select one member per team to submit the assignment via Moodle.

Keep in mind that all your other assignments as well as the project will heavily rely on this assignment.